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Problem statement: Represent a given graph using adjacency matrix/list to perform DFS and using adjacency list to perform BFS. Use the map of the area around the college as the graph. Identify the prominent land marks as nodes and perform DFS and BFS on that.

```
from collections import deque

landmarks = ["College", "Library", "Canteen", "Auditorium", "Playground",
"Hostel"]
edges = [(0,1), (0,2), (1,3), (2,4), (4,5)]

# Adjacency Matrix
matrix = [[0]*6 for _ in range(6)]
for u, v in edges: matrix[u][v] = matrix[v][u] = 1

# Adjacency List
adj = [[] for _ in range(6)]
for u, v in edges: adj[u].append(v); adj[v].append(u)

def dfs_matrix(v, vis):
    vis[v] = True
    print(landmarks[v], end=" → ")
    for i in range(6):
        if matrix[v][i] and not vis[i]:
            dfs_matrix(i, vis)

def bfs_list(start):
    vis = [False]*6
    q = deque([start])
    vis[start] = True
    while q:
        v = q.popleft()
        print(landmarks[v], end=" → ")
        for u in adj[v]:
            if not vis[u]:
                vis[u] = True
                q.append(u)
```

OUTPUT:

```
DFS (Matrix):
College → Library → Auditorium → Canteen → Playground → Hostel →
BFS (List):
College → Library → Canteen → Auditorium → Playground → Hostel →
PS C:\Users\Manav\Desktop\exp of DSA>
```